

OSI Layers Used When Sending a WhatsApp Message

When I send a message using WhatsApp, different OSI layers work together to transfer the message from my phone to the receiver's phone.

1. Application Layer (Layer 7)

This layer allows the WhatsApp application to use network services. When I type a message and press the **Send** button, the Application Layer starts the communication process.

Example: I type "Hello" in WhatsApp and send it to my friend.

2. Presentation Layer (Layer 6)

This layer handles data formatting and encryption. Before the message is sent over the network, the data is prepared in a suitable format and protected using encryption.

Example: My "Hello" message is encrypted so that unauthorized people cannot easily read it while it travels across the network.

3. Transport Layer (Layer 4)

This layer manages end-to-end data transfer between devices. Transport protocols help deliver application data across the network.

Example: The message data is transported from my phone toward WhatsApp's service and eventually to the receiver.

4. Network Layer (Layer 3)

This layer uses IP addressing and routing to move packets across different networks.

Example: Routers select paths to send the message packets from my Wi-Fi or mobile network through the internet toward the destination.

5. Physical Layer (Layer 1)

This layer converts data bits into physical signals such as radio, electrical, or optical signals.

Example: If I use mobile data or Wi-Fi, the message is transmitted partly as radio signals between my phone and nearby network equipment.